Measurement System Analysis (Type 1 Gauge Study)

Battery Cycling Quality Analytics — Measure Phase

Objective

Validate that the capacity measurement computed from raw current and time logs (trapezoidal integration of discharge current) agrees with the NASA Prognostics Center of Excellence reported discharge capacity. Measurement validity is a pre-condition for all downstream Statistical Process Control, capability, and reliability work.

Method

Type 1 Gauge Study. One deterministic measurement method (re-derived capacity) compared against one authoritative reference (NASA reported capacity). Acceptance threshold: overall Mean Absolute Percentage Error (MAPE) ≤ 1.0 percent across all 4 cells (B0005, B0006, B0007, B0018) and all cycles. Secondary check: Bland-Altman limits of agreement (mean difference ± 1.96 × standard deviation).

Results — per cell and overall

Cell	N cycles	Mean NASA (Ah)	Mean computed (Ah)	Mean residual (Ah)	MAPE (%)	Max |% error|
B0005	168	1.5725	1.5759	+0.0034	0.213	0.318
B0006	168	1.5469	1.5607	+0.0138	0.942	1.633
B0007	168	1.6444	1.6659	+0.0215	1.328	1.722
B0018	132	1.5577	1.5714	+0.0137	0.896	1.453
ALL	**636**	**1.5817**	**1.5947**	**+0.0131**	**0.842**	**1.722**

Overall MAPE: 0.842% | Threshold: ≤ 1.0% | **Verdict: PASS**

Interpretation

The re-derived capacity measurement agrees with the NASA reference to within 0.84 percent on average across all 4 cells and all cycles. Residuals show no systematic drift across cycle life (see chart Panel B) and the Bland-Altman limits of agreement (Panel C) are tight relative to the measured range of capacity. The measurement system is accepted for use in all downstream Statistical Process Control, Cp / Cpk capability, and Weibull reliability analyses.

Measurement system verdict

Accepted. No systemic bias detected. Measurement variation is small relative to the cell-to-cell and cycle-to-cycle variation that will be analyzed in subsequent DMAIC phases.

